

TicketMinds — AI-Powered Multilingual Customer Support Platform

Introduction

TicketMinds is an AI-powered multilingual customer support platform designed to eliminate communication barriers between customers and support engineers.

Problem Statement

Customers often speak different languages while support engineers primarily communicate in English, creating communication barriers and delayed issue resolution.

Proposed Solution

The system automatically detects customer language, translates messages into English for engineers, and translates engineer responses back into the customer's language.

Objectives

- Detect customer language automatically
- Translate messages in real time
- Enable multilingual communication
- Build role-based dashboards
- Improve customer support efficiency

Key Features

- Login and Signup
- Client and Engineer dashboards
- Real-time translation
- AI-powered sentiment analysis
- Ticket management system

System Workflow

1. Client sends message in native language.
2. System detects language.
3. Message translated into English.
4. Engineer replies in English.
5. Response translated back to client language.

Advantages

- Eliminates language barriers
- Faster ticket resolution
- Improved customer satisfaction
- AI-powered support workflow

Future Enhancements

- Voice-to-text support
- Mobile application
- Real-time notifications
- AI-generated replies

Conclusion

TicketMinds demonstrates how Artificial Intelligence and Natural Language Processing can improve multilingual customer support systems.

Technologies Used

Component	Technology
Frontend	Streamlit
Backend	Python
Database	SQLite
Translation	deep-translator
Language Detection	langdetect
AI Model	Ollama + Llama 3
Version Control	GitHub
Deployment	Streamlit Cloud